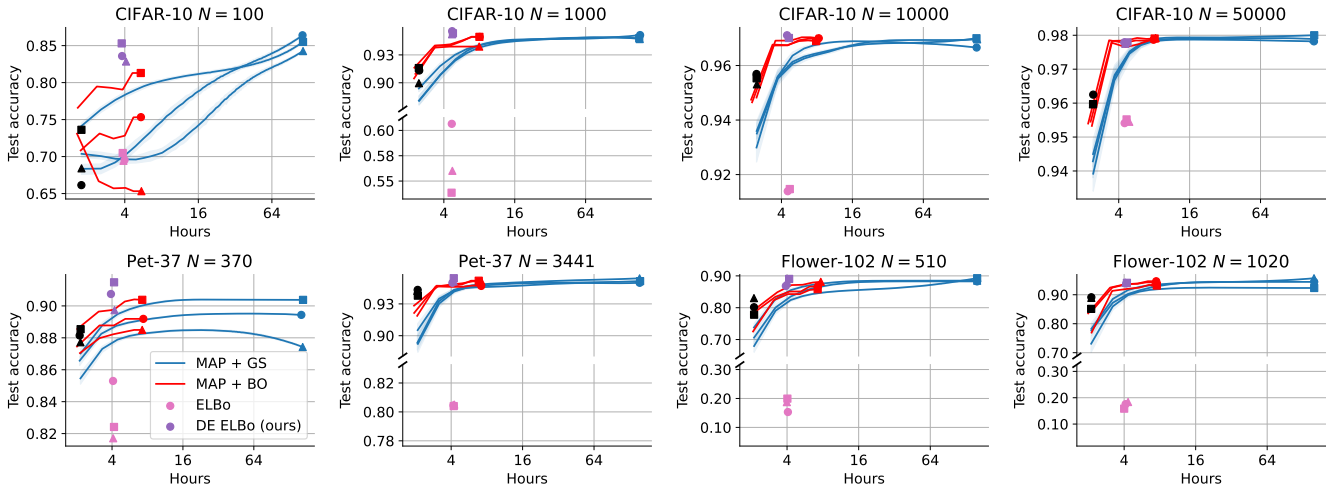

Figure 3: Test-set accuracy on CIFAR-10, Pet-37, and Flower-102 over total runtime for L2-SP with *MAP + grid search (GS)*, *MAP + Bayesian optimization (BO)*, ELBo, and our *data-emphasized ELBo* (DE ELBo) using ConvNeXt-Tiny (see ViT-B/16 in Fig. 5 and ResNet-50 in Fig. 6). We run each method on 3 separate training sets of size N (3 different marker styles). **Takeaway: Our DE ELBo achieves as good or better performance at small dataset sizes and similar performance at large dataset sizes with far less compute time.** To make the blue curves, we did the full grid search once (markers). Then, at each given shorter compute time, we subsampled a fraction of all hyperparameter configurations with that runtime and choose the best via validation NLL. Averaging this over 500 subsamples at each runtime created each blue line. To make the red curves, we use BO to select candidate hyperparameter configurations and choose the best via validation NLL. Averaging this over 5 BO runs created each red line. Black markers are $\alpha = \beta = 0.0001$ and the learning rate from PyTorch's *Transfer Learning for Computer Vision Tutorial*: https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html.